

Z-test

A test used to compare the sample mean to the population mean.

--> Calculating the Z-score tell us how much the sample mean is different from the population mean by looking at how much the data normally varies.

Use when

-sample size is large >30

-Standard deviation is known

$$\text{Z-Score} = \frac{\bar{x} - \mu}{\sigma}$$

- $\bar{x}$: mean of the sample.
- μ : mean of the population.
- σ : Standard deviation of the population

Parametric test are statistical methods that make assumption that the data follows normal distribution.

<u>Z-test</u>	<u>t-test</u>	<u>F-test</u>
Tests if a sample mean differs from a known population mean.	Compares means when population standard deviation is unknown.	Compares variances of two or more groups.
Population standard deviation is known and sample size is large.	Small samples or unknown population standard deviation.	To test if group variances are significantly different.
One-Sample Test: $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$ Two-Sample Test: $Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	One- sample: $t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$ Two-Sample Test: $t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ Paired t-Test: $t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$ d= difference	$F = \frac{s_1^2}{s_2^2}$

Critical value = significance level / 2

-if the z-test > Critical Value

we reject the null hypothesis

-if the z-test < Critical Value

we do not have enough evidence to reject the null hypothesis

P-value

If $p \leq \alpha$: reject H_0

If $p > \alpha$: fail to reject H_0